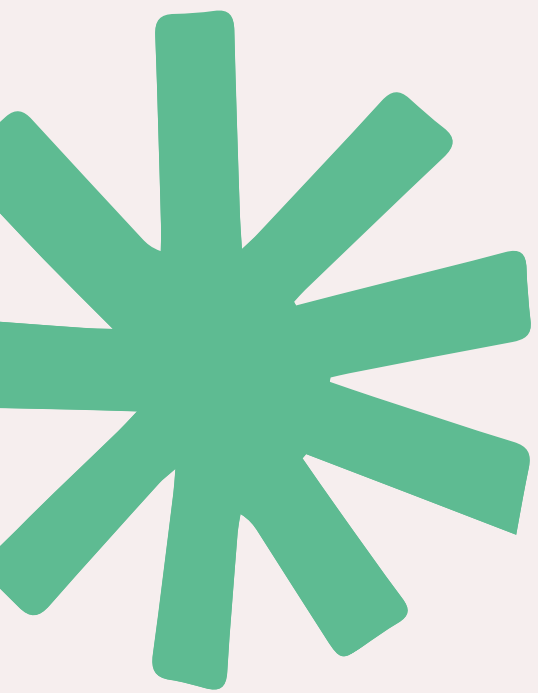




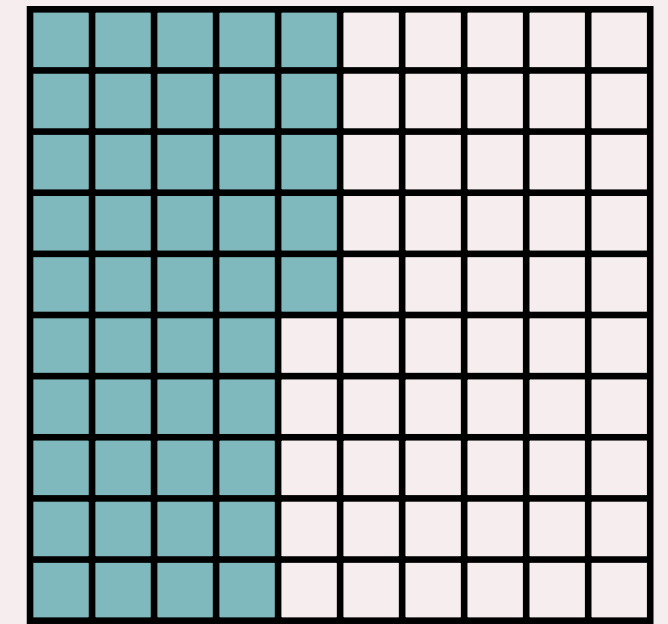
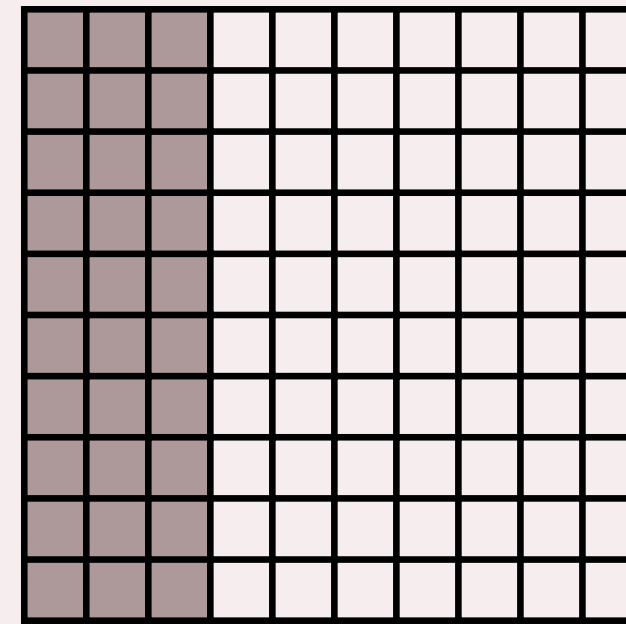
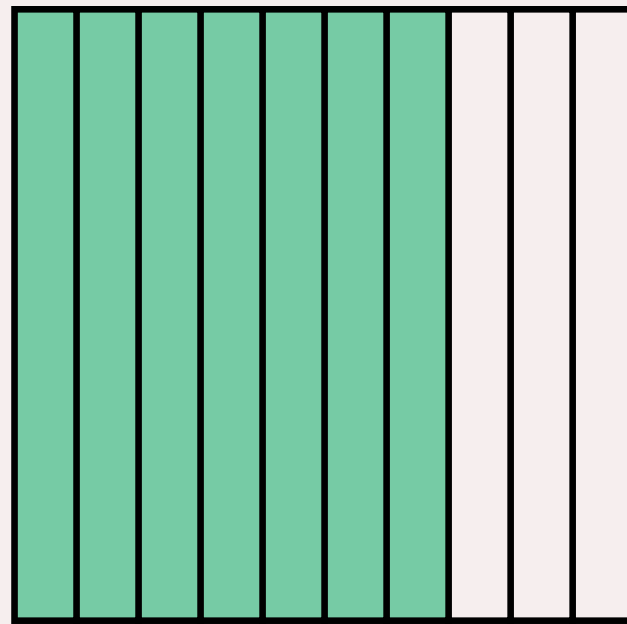
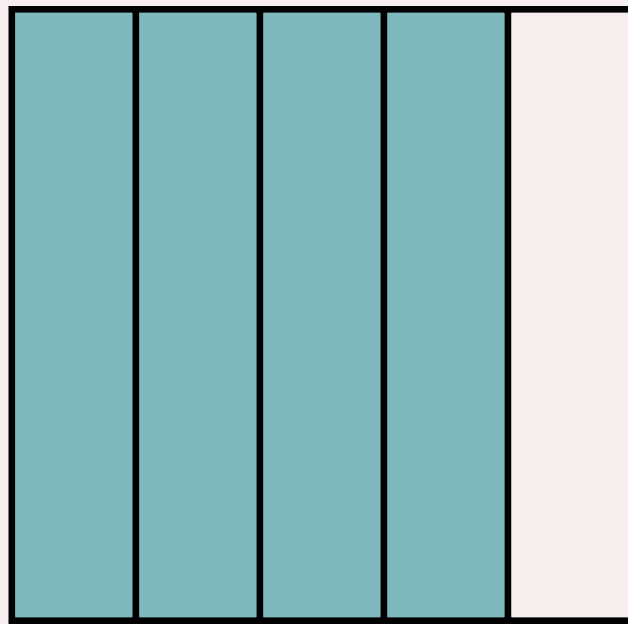
Converting between

Fractions, Decimals and Percentages



Fractions

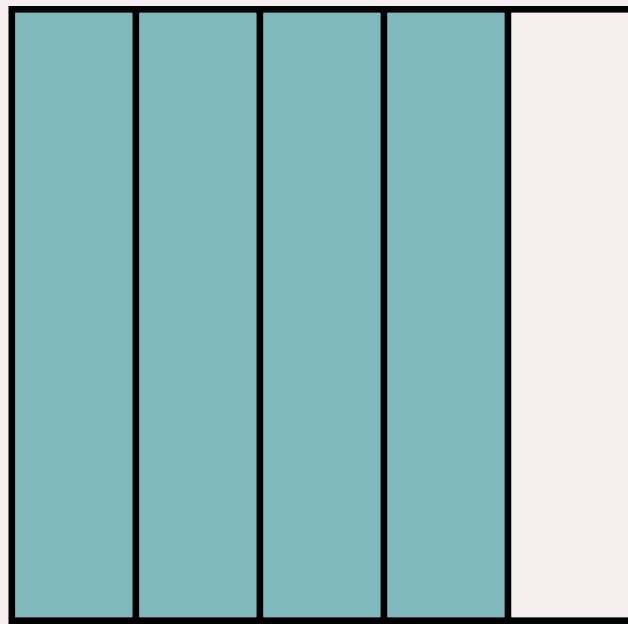
A fraction represents a part of a whole.
Write down the fractions represented by the diagrams:



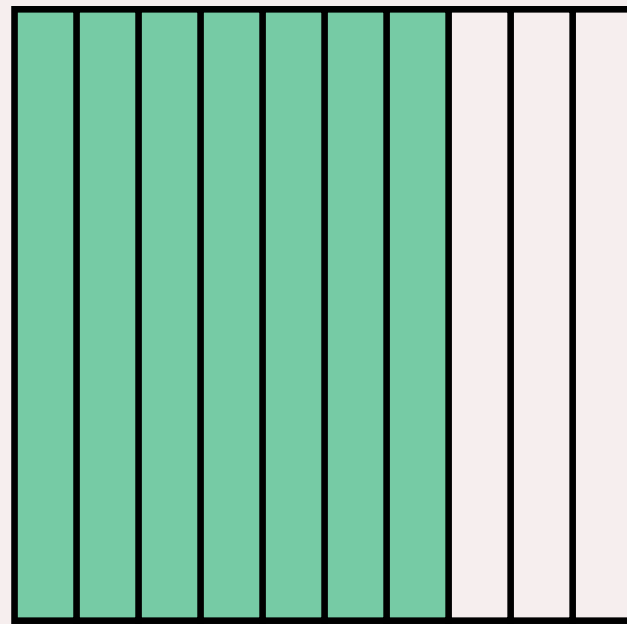
Fractions

Answer Key

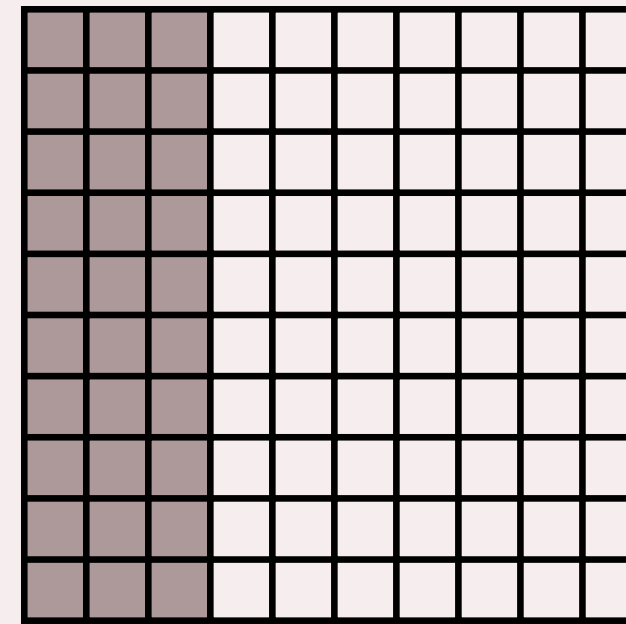
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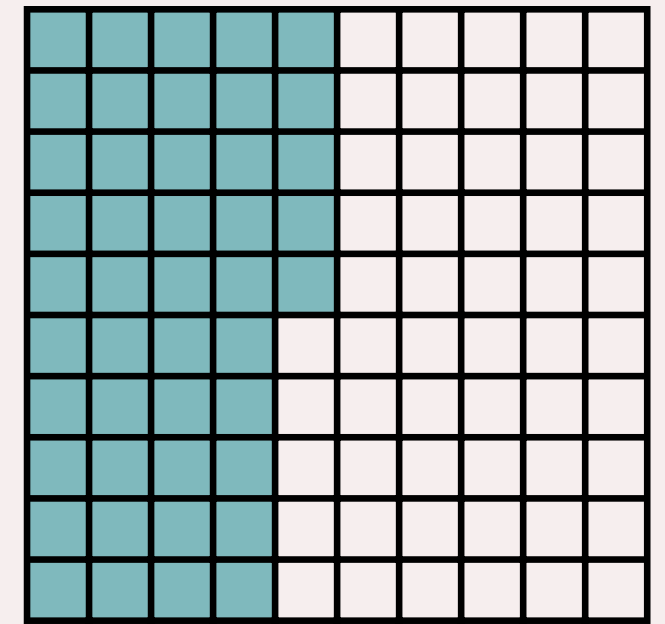
$$\frac{4}{5}$$



$$\frac{7}{10}$$



$$\frac{3}{10}$$

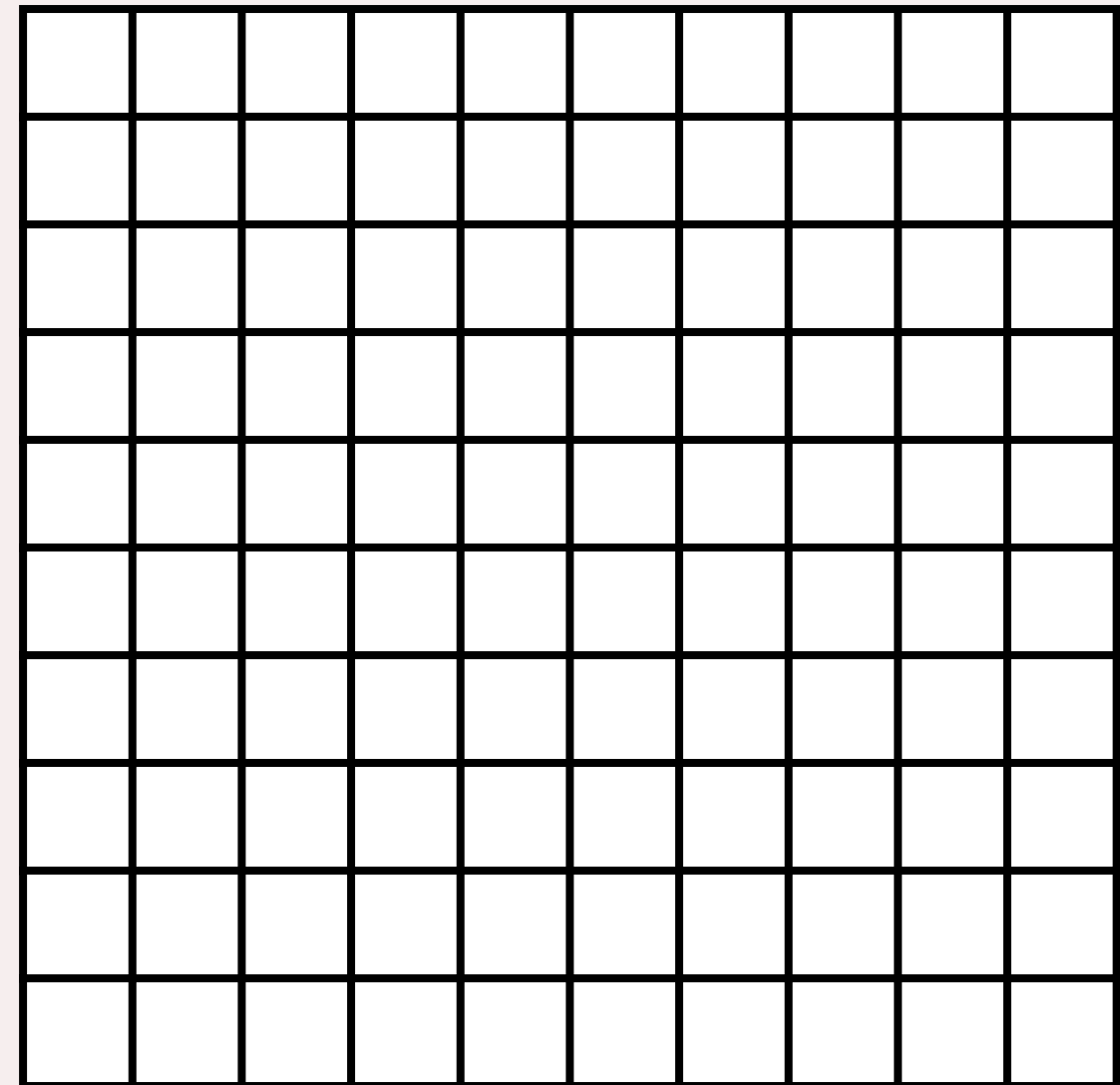
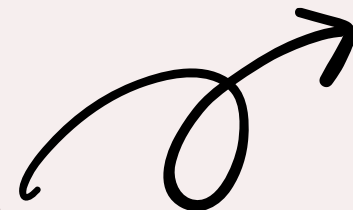


$$\frac{45}{100}$$

Percentages

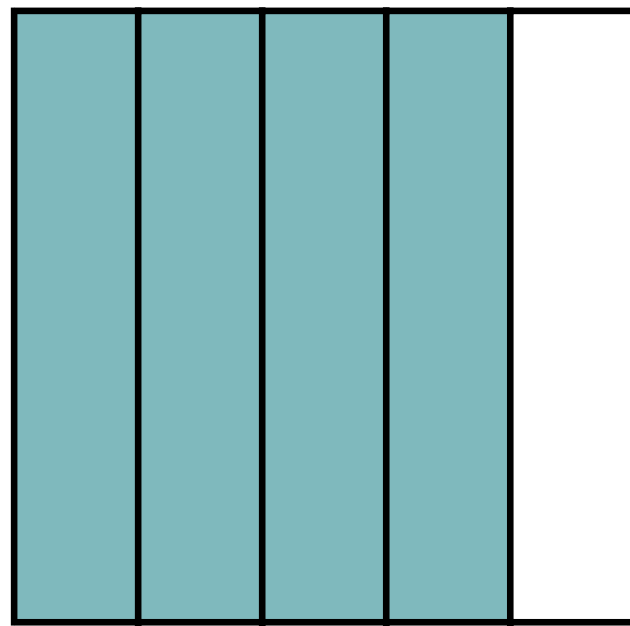
Percentages are a way of expressing a part of a whole as a fraction of 100. They are useful for comparing quantities

Out of 100



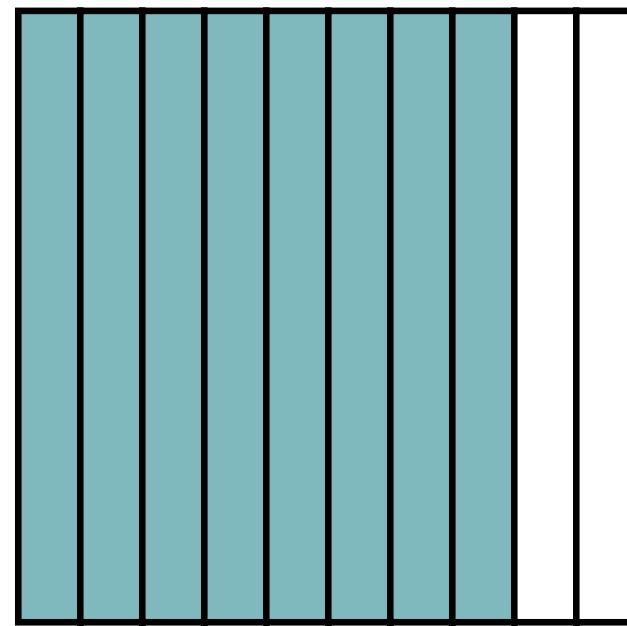
Fractions to Percentages

In order to convert a fraction into a percentage, you need to find the equivalent where the denominator is 100.



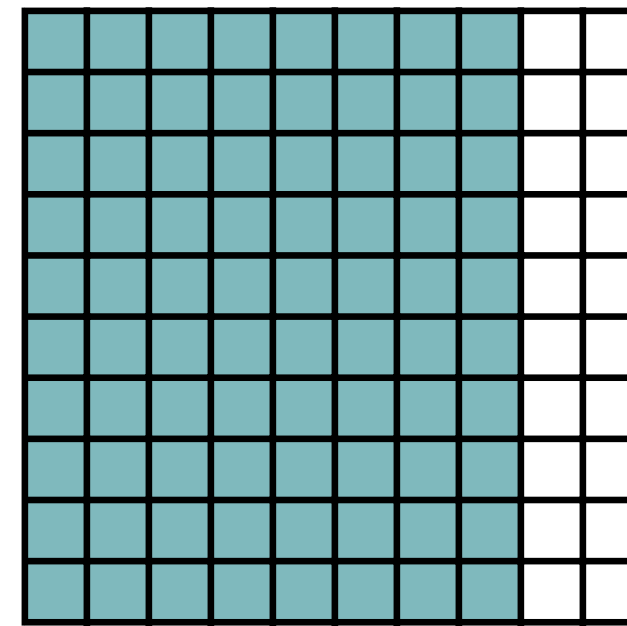
$$\frac{4}{5}$$

=



$$\frac{8}{10}$$

=



$$\frac{80}{100}$$

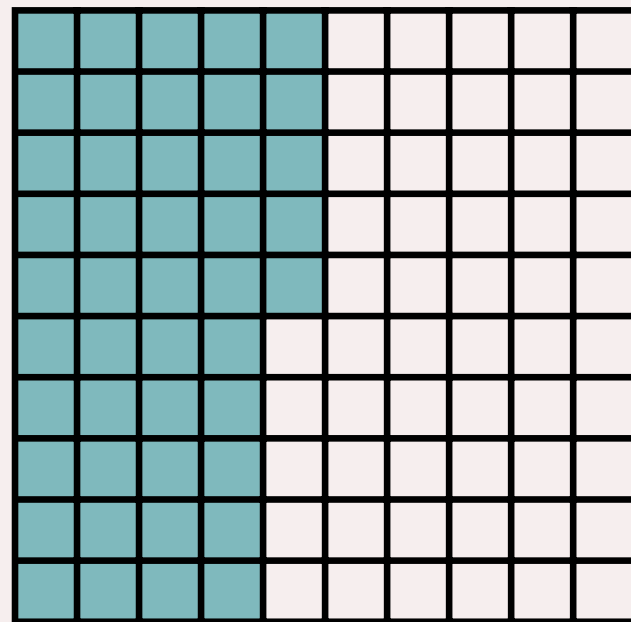
→

80%

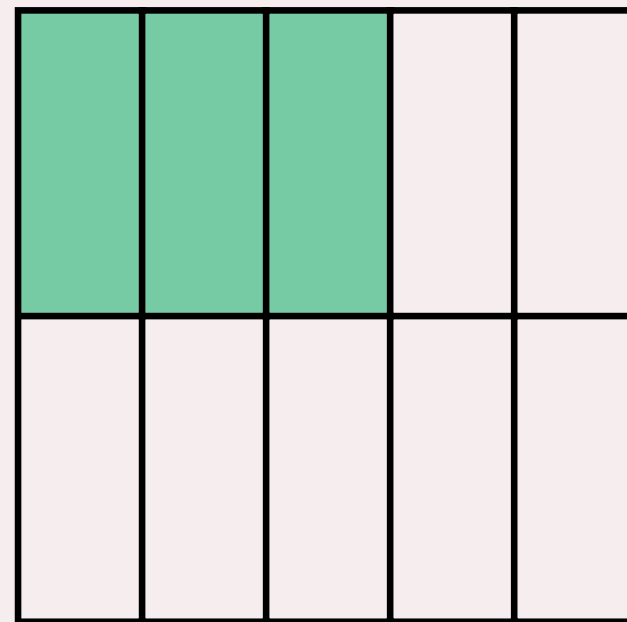
The numerator is the percentage.

Fractions to Percentages

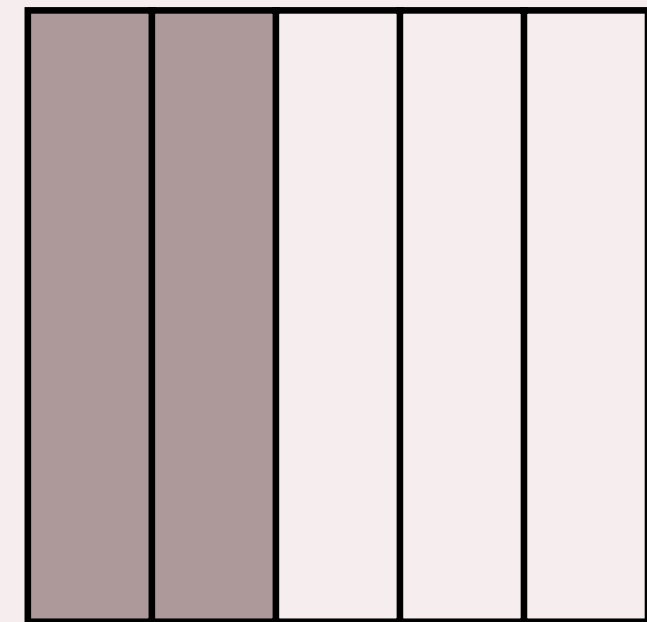
Convert the following fractions into percentages:



$$\frac{45}{100} =$$



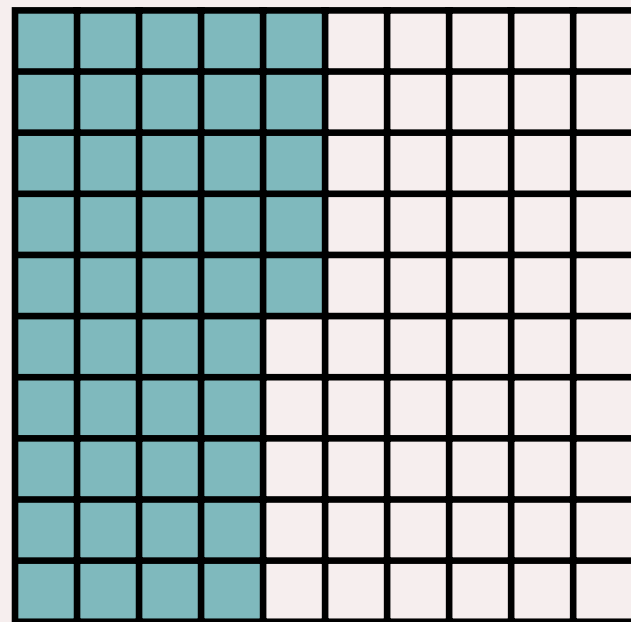
$$\frac{3}{10} =$$



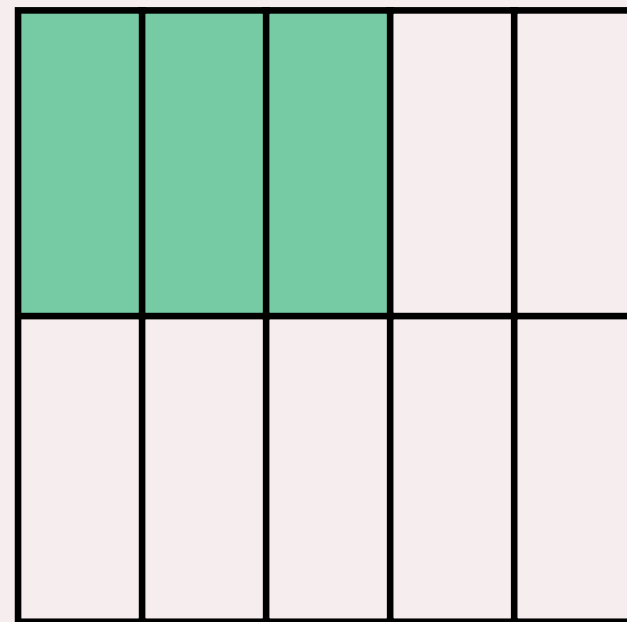
$$\frac{2}{5} =$$

Fractions to Percentages

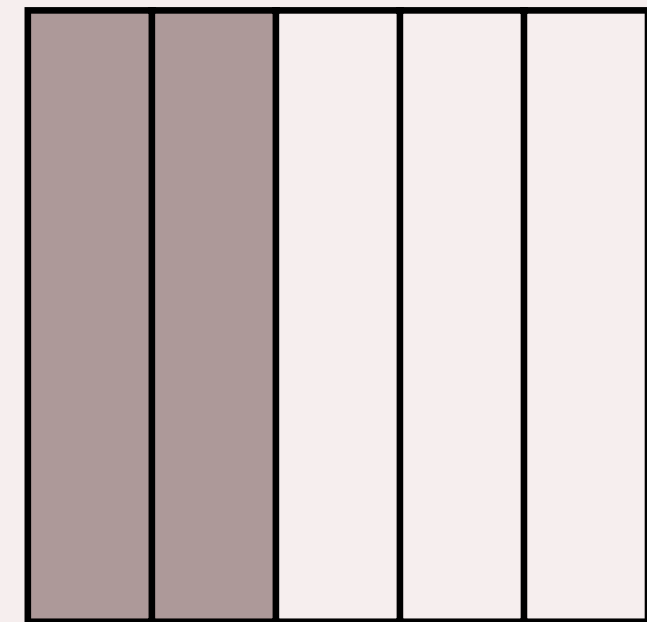
Answer Key: Convert the following fractions into percentages:



$$\frac{45}{100} = 45\%$$



$$\frac{3}{10} = \frac{30}{100} = 30\%$$



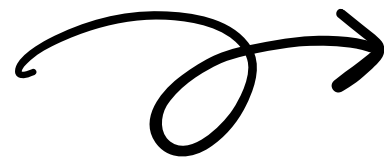
$$\frac{2}{5} = \frac{40}{100} = 40\%$$

Percentages to Decimals

In order to convert percentages into decimals, divide by 100

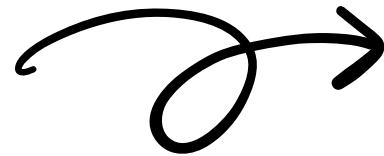
Divide by 100

88%



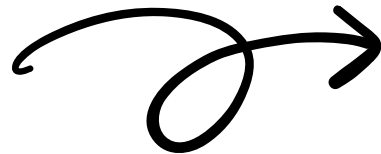
0.88

70%



0.7

7%



0.07

Fractions to Decimals

To convert fractions into decimals,
divide the numerator by the denominator

$$\frac{3}{8} = 3 \div 8 = 0.375$$

$$\begin{array}{r} 0.375 \\ 8 \overline{) 3.000} \end{array}$$

Divide the numerator by the denominator

Fractions, Decimals, Percentages

Fraction	Decimal	Percentage
$\frac{1}{5}$		
	0.05	
		44%
	0.68	
$\frac{5}{8}$		

Fractions, Decimals, Percentages

Answer Key

Fraction	Decimal	Percentage
$\frac{1}{5}$	0.2	20%
$\frac{5}{100} = \frac{1}{20}$	0.05	5%
$\frac{44}{100} = \frac{22}{50} = \frac{11}{25}$	0.44	44%
$\frac{68}{100} = \frac{34}{50} = \frac{17}{25}$	0.68	68%
$\frac{5}{8}$	0.625	62.5%



Great Job!

Add decimals to hundredths

Just like adding whole numbers, we can add decimals by lining up the digits based on their place value.

Let's add 2.18 and 1.31.

Units	.	Tenths	Hundredths
2	.	1	8
+ 1	.	3	1



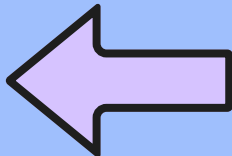
TIP:

When trying to line up the values, you can line up the decimal symbol first.

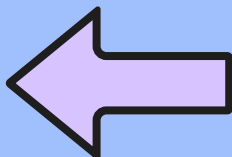
Add decimals to hundredths

	Units	.	Tenths	Hundredths
	2	.	1	8
+	1	.	3	1
	3	.	4	9

After lining up the digits of 2.18 and 1.31 based on place value, we can now add from right to left.



Start by adding the values on the Hundredths place, then the Tenths place, and finally the Units place.



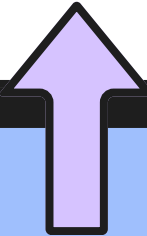
Remember to write the decimal symbol in the sum!

The sum of 2.18 and 1.31 is **3.49**.

Let us try
another
example!

Add 3.46 and 2.72.

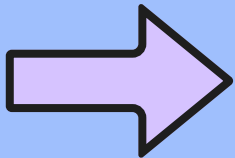
Units	.	Tenths	Hundredths
3	.	4	6
+ 2	.	7	2
11			8



But wait! The sum of 4-tenths and 7-tenths is 11-tenths. When we get a two-digit sum, we need to perform **regrouping**.

Add decimals to hundredths

We will regroup 10-tenths as 1 whole number or 1 unit.



	Units	.	Tenths	Hundredths
	1			
	3	.	4	6
+	2	.	7	2
	6	.	1	8

The sum of 3.46 and 2.72 is **6.18**.

Subtract decimals to hundredths

Just like in adding decimals,
we follow the same method in
subtraction.

Let us do $8.74 - 2.5$.

Units	.	Tenths	Hundredths
8	.	7	4
- 2	.	5	0
6	.	2	4

$$8.74 - 2.5 = 6.24$$



TIP:

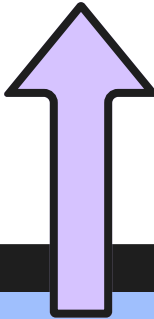
After lining up the digits in their place values,
any empty space can be filled in with zeroes.

2.5 is exactly the same as 2.50.

Subtract decimals to hundredths

Let us try $23.15 - 10.42$.

Tens	Units	.	Tenths	Hundredths
2	3	.	1	5
- 1	0	.	4	2
				3

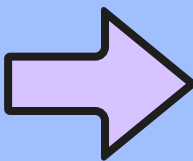


But wait! How do we subtract 4-tenths from 1-tenths if it is bigger? Since we cannot subtract directly in the tenths column, we'll do **regrouping** of 1 from the units column.

Subtract decimals to hundredths

Let us try $23.15 - 10.42$.

Tens	Units	.	Tenths	Hundredths
2	3 2	.	1 11	5
- 1	0	.	4	2
				3



Tens	Units	.	Tenths	Hundredths
2	2	.	11	5
- 1	0	.	4	2
1	2	.	7	3

Take away 1 from the units place, and it will be added to the tenths place as 10-tenths.

$$23.15 - 10.42 = 12.73$$

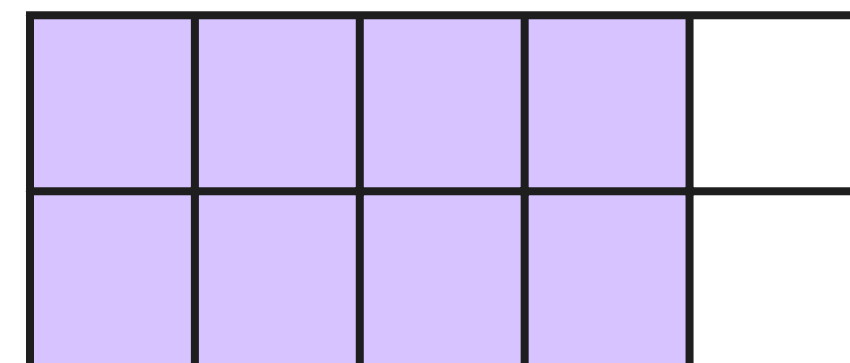
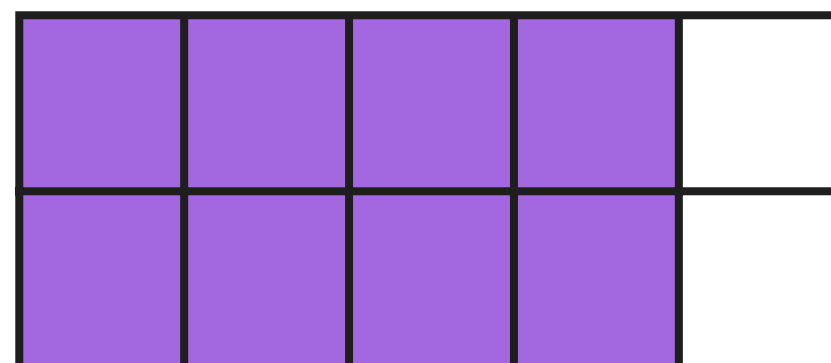
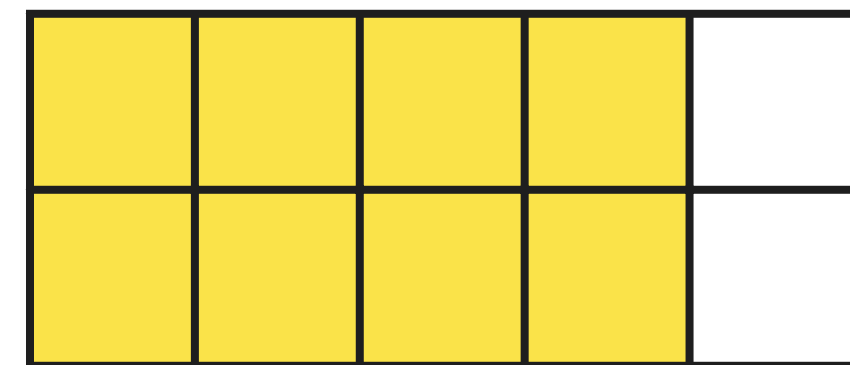
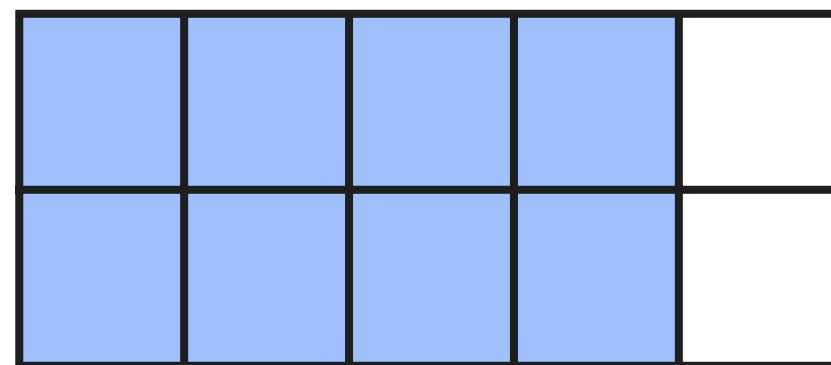
Multiplying decimals

There are multiple ways we can multiply decimals.

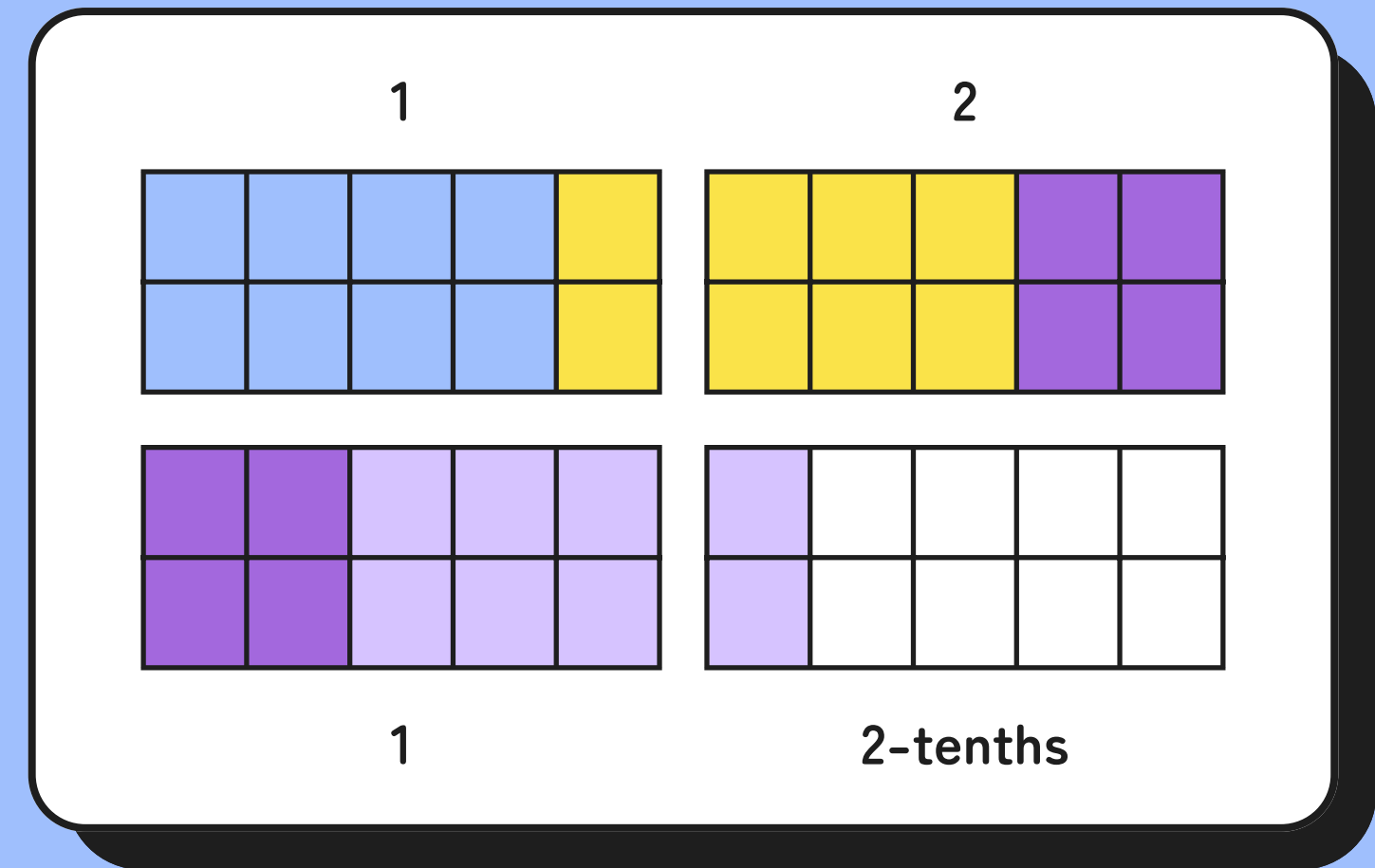
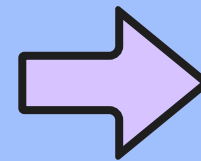
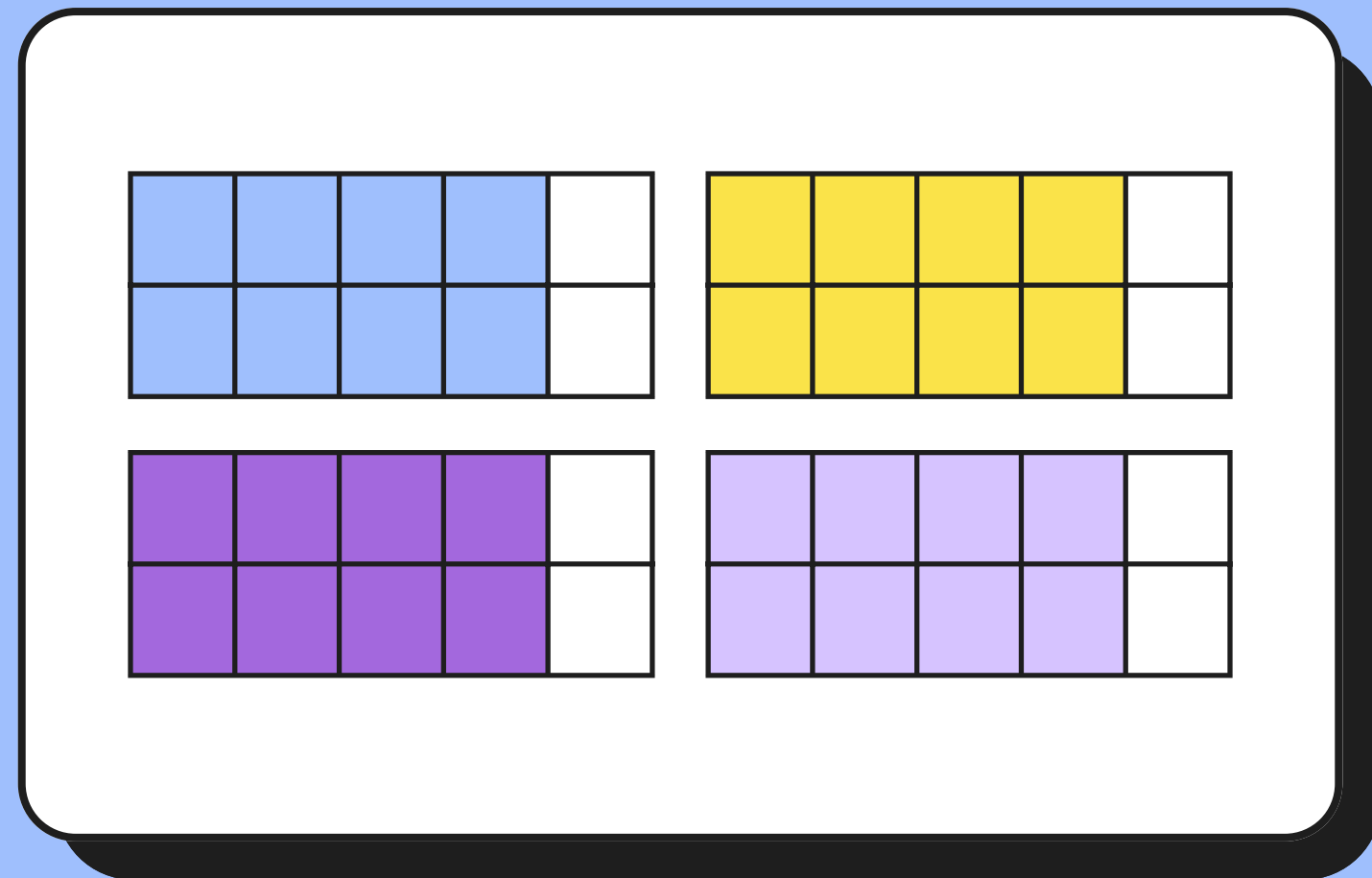
Example:

$$0.8 \times 4$$

Using visual models, we have 4 groups of 0.8



Multiplying decimals by a whole number



We will move and fill in any empty boxes to combine the models.

After moving the colored boxes, we see that it fills in 3 whole models and 2-tenths of the last model.

That means that:

$$0.8 \times 4 = \mathbf{3.2} \text{ or } \mathbf{3 \text{ and } 2\text{-tenths}}$$

How about for 1.36×8 ?

- 1 Use a rectangular area model to multiply the values.

	1	0.3	0.06
8	8	2.4	0.48

$$1.36 \times 8 = 8 + 2.4 + 0.48 = 10.88$$

- 2 Write the numbers in a place value table. Add the values in each column.

	Ones	.	Tenths	Hundredths
	8	.	0	0
	2	.	4	0
+	0	.	4	8
	1	0	.	8

$$1.36 \times 8 = 10.88$$

Multiplying decimals by 2 digit numbers

Another method we can use is the standard algorithm when multiplying numbers.

Example: 8.15×12

- 1 Write the numbers vertically.
- 2 Multiply 8.15×2 .
- 3 Multiply 8.15×1 .
- 4 Add the products.
- 5 Move the **decimal point 2 times to the left** since 8.15 has 2 decimal places.

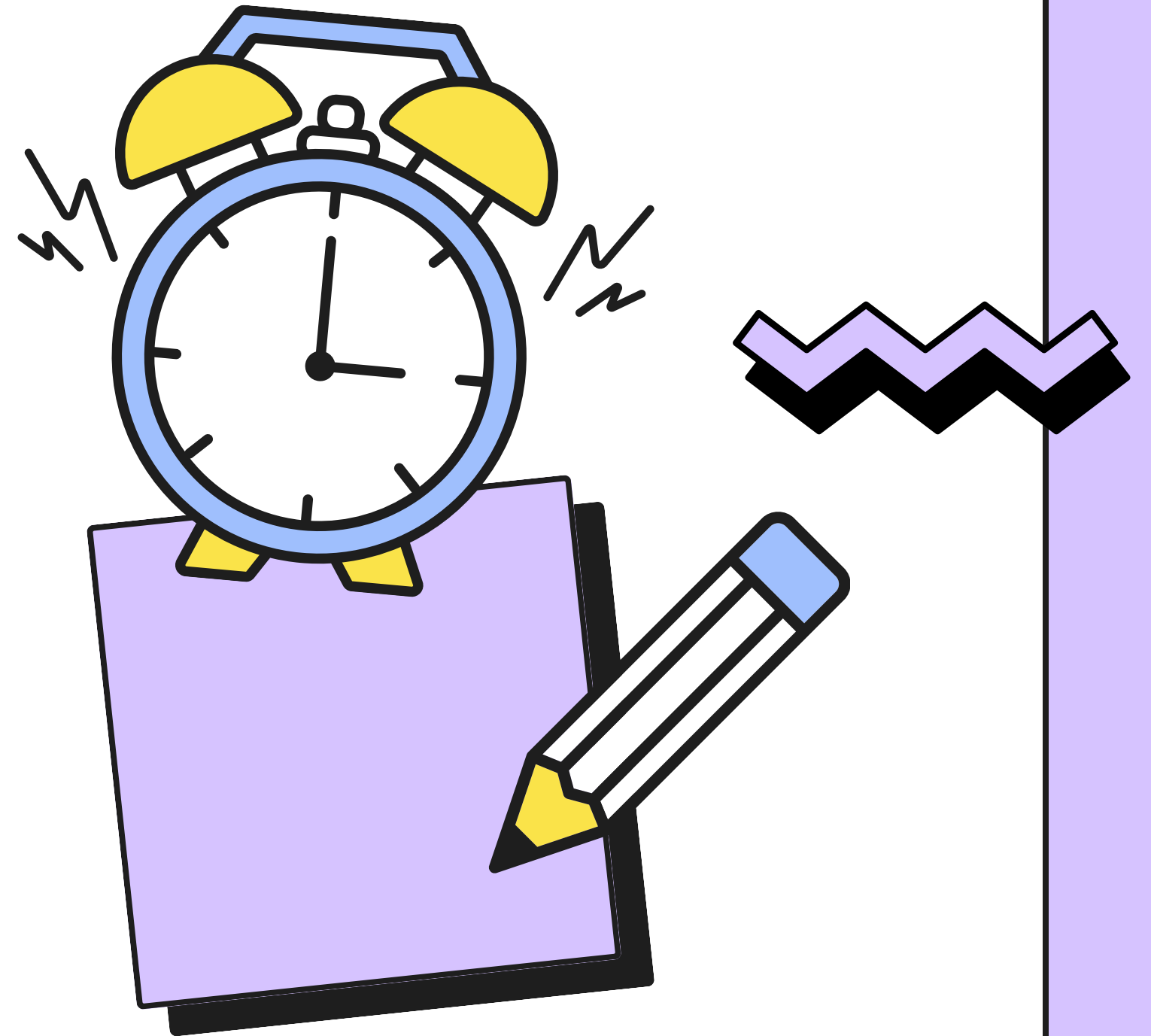
$$\begin{array}{r} 8.15 \\ \times 12 \\ \hline 16\ 30 \\ 81\ 50 \\ \hline 97\ 80. \end{array}$$

Answer:
 $8.15 \times 12 = 97.80$

Mid-Discussion Break

Divide 175 by 4.

Use the process of long division.



Answer Key

Mid-Discussion Break

Divide 175 by 4.

Use the process of long division.

$$\begin{array}{r} 43 \text{ r.}3 \\ 4 \overline{) 175} \\ \underline{-16} \\ 15 \\ \underline{-12} \\ 3 \end{array}$$



NOTE:

In our previous lessons, when we can no longer divide the remaining value, we stop and call it the remainder.

Division where answers are decimals

Let's continue, but instead of having a remainder 3, we will divide further to get the decimal numbers.

 **NOTE:** Adding decimal 0 to a number does not change the number's value.

- ① Add **0 tenths**.
3 will become **30-tenths**.
- ② Divide normally to get 7.
- ③ Add **0 hundredths**.
2 will become **20-hundredths**.
- ④ Divide normally to get 5.

$$\begin{array}{r} 43.75 \\ 4 \overline{) 175.00} \\ \underline{-16} \\ 15 \\ \underline{-12} \\ 30 \\ \underline{-28} \\ 20 \\ \underline{-20} \\ 0 \end{array}$$

Answer:
 $175 \div 4 = 43.75$

Dividing decimals by a whole number

The method is very similar to dividing whole numbers.

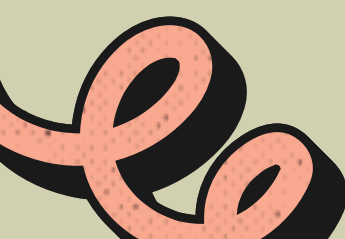
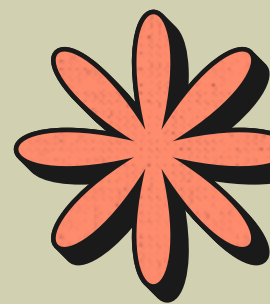
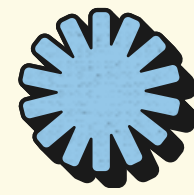
Example: $16.28 \div 2$

- ① **Place the decimal point** in the quotient directly above the decimal point of the dividend.
- ② **Divide normally**, starting from the left, and ignore the decimal for now.
- ③ Now you've got your answer.

$$\begin{array}{r} 8.14 \\ 2 \overline{) 16.28} \\ \underline{- 16} \\ 02 \\ \underline{- 2} \\ 08 \\ \underline{- 8} \\ 0 \end{array}$$

Answer:

$$16.28 \div 2 = 8.14$$



Thank you very much!

For everyone who has done
a great job...

See you next time!